Math 54 — Week 6

Linearly independent sets and bases; dimension and rank (Lay 4.3, 4.5)

§1 Facts used from Weeks 1–5

(W1) (Week 3) A set of two or more vectors is dependent exactly when one of them is a combination of the others. More than n vectors in $\mathbb{R}^n$ are dependent.

(W2) (Weeks 2–3) The columns of A are independent $\iff$ a pivot in every column; they span $\mathbb{R}^m$ $\iff$ a pivot in every row. A homogeneous system with more unknowns than equations has a nontrivial solution.

(W3) (Week 1) Row operations do not change the solution set of $A\mathbf{x} = \mathbf{0}$, and each can be undone by a row operation.

(W4) (Week 5) Subspaces; a span is a subspace; $\operatorname{Nul} A$, $\operatorname{Col} A$, $\operatorname{Row} A = \operatorname{Col} A^T$. The parametric vector form of $A\mathbf{x} = \mathbf{0}$ gives a linearly independent spanning set for $\operatorname{Nul} A$, one vector per free variable.

(W5) (Weeks 4–5) Invertible matrix theorem, including $\det A \neq 0$. Each pivot lies in a different row and column, so an $m \times n$ matrix has at most $\min(m, n)$ pivots.

§2 This week

§2.1 Linearly independent sets; bases (Lay 4.3)

The question. The five columns of the matrix below span $\operatorname{Col} A$, but some of them are redundant. Which can be dropped, what is the smallest kind of set that still spans, and how do we find one for $\operatorname{Nul} A$, $\operatorname{Col} A$ and $\operatorname{Row} A$?

$$A = \begin{bmatrix} 1 & 4 & 0 & 2 & -1 \\ 3 & 12 & 1 & 5 & 5 \\ 2 & 8 & 1 & 3 & 2 \\ 5 & 20 & 2 & 8 & 8 \end{bmatrix}$$

Definition. A set $\{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$ in a vector space V is **linearly independent** if $c_1\mathbf{v}_1 + \cdots + c_p\mathbf{v}_p = \mathbf{0}$ only for $c_1 = \cdots = c_p = 0$, and **linearly dependent** otherwise.

Theorem (Characterization of dependent sets (Lay 4.3, Thm. 4))
An indexed set $\{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$ of two or more vectors, with $\mathbf{v}_1 \neq \mathbf{0}$, is linearly dependent if and only if some $\mathbf{v}_j$ with $j > 1$ is a linear combination of the preceding vectors $\mathbf{v}_1, \ldots, \mathbf{v}_{j-1}$.

Proof. 1. **Assume $\mathbf{v}_j = c_1\mathbf{v}_1 + \cdots + c_{j-1}\mathbf{v}_{j-1}$.** Then $c_1\mathbf{v}_1 + \cdots + c_{j-1}\mathbf{v}_{j-1} - \mathbf{v}_j = \mathbf{0}$, with weight $-1 \neq 0$ on $\mathbf{v}_j$ (and weight 0 on the vectors after $\mathbf{v}_j$): a dependence relation.

2. **Assume the set is dependent**, with $c_1\mathbf{v}_1 + \cdots + c_p\mathbf{v}_p = \mathbf{0}$, weights not all 0. Let j be the largest index with $c_j \neq 0$. If $j = 1$, then $c_1\mathbf{v}_1 = \mathbf{0}$ with $c_1 \neq 0$, so $\mathbf{v}_1 = \frac{1}{c_1}\mathbf{0} = \mathbf{0}$, which is false. So $j > 1$, and solving for $\mathbf{v}_j$ gives

$$\mathbf{v}_j = -\frac{1}{c_j}\left(c_1\mathbf{v}_1 + \cdots + c_{j-1}\mathbf{v}_{j-1}\right).$$

□

The facts from Week 3 hold in any vector space with the same proofs, since those proofs use only the vector space axioms: a set containing $\mathbf{0}$ is dependent ($1 \cdot \mathbf{0} = \mathbf{0}$ is a relation with a nonzero weight), and two vectors are dependent exactly when one is a multiple of the other.

Examples in function spaces.

- In $\mathbb{P}_n$, $\{1, t, \ldots, t^n\}$ is independent: two polynomials are equal exactly when their coefficients are equal, so $c_0 + c_1 t + \cdots + c_n t^n$ is the zero polynomial only if all $c_i = 0$. (Viewed as functions on $\mathbb{R}$, the same holds because a nonzero polynomial of degree at most n has at most n zeros; that fact from algebra is cited.)
- **Polynomials of different degrees are independent** (none being the zero polynomial). In a relation $c_1 p_1 + \cdots + c_k p_k = 0$ with some $c_i \neq 0$, let p_i have the largest degree d among those with $c_i \neq 0$. No other polynomial with a nonzero weight has degree d, so the coefficient of t^d in the sum is c_i times the leading coefficient of p_i, which is not 0. So the sum is not the zero polynomial, and every weight must be 0.
- $\{\sin t \cos t,\ \sin 2t\}$ is dependent in the space of functions on $\mathbb{R}$, since $\sin 2t = 2 \sin t \cos t$. $\{\sin t, \cos t\}$ is independent: if $\sin t = c \cos t$ for all t, then $t = 0$ gives $0 = c$, and then $\sin t = 0$ for all t, which is false; if $\cos t = c \sin t$ for all t, then $t = 0$ gives $1 = 0$.

Definition. A **basis** for a subspace H of V is a set $\mathcal{B} = \{\mathbf{b}_1, \ldots, \mathbf{b}_p\}$ in H that is linearly independent and spans H.

- The columns $\mathbf{e}_1, \ldots, \mathbf{e}_n$ of I_n form the **standard basis** of $\mathbb{R}^n$. $\{1, t, \ldots, t^n\}$ is the standard basis of $\mathbb{P}_n$: it spans by definition of $\mathbb{P}_n$ and is independent by the example above.
- The columns of an invertible $n \times n$ matrix form a basis of $\mathbb{R}^n$: by the invertible matrix theorem they are independent and span $\mathbb{R}^n$.

Theorem (The spanning set theorem (Lay 4.3, Thm. 5))

Let $S = \{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$ be a set in V and $H = \operatorname{Span} S$.

(a) If some $\mathbf{v}_k$ in S is a combination of the other vectors of S, then removing $\mathbf{v}_k$ from S does not change the span.

(b) If $H \neq \{\mathbf{0}\}$, some subset of S is a basis for H.

Proof. 1. (a) Write $\mathbf{v}_k = \sum_{i \neq k} a_i \mathbf{v}_i$. In any combination $\sum_i c_i \mathbf{v}_i$, substitute this for $\mathbf{v}_k$: the result $\sum_{i \neq k} (c_i + c_k a_i) \mathbf{v}_i$ uses only the other vectors. So every vector of H lies in the span of S without $\mathbf{v}_k$, and that span is contained in H.

2. (b) If S is independent, it is a basis of H. If S is dependent and has two or more vectors, then by (W1) some vector of S is a combination of the others; remove it, which by (a) leaves the span equal to H.

3. Repeat step 2. Each removal leaves a smaller set that still spans H. If only one vector is left, it spans $H \neq \{\mathbf{0}\}$, so it is not $\mathbf{0}$, and one nonzero vector is independent. So the process stops at an independent subset of S that spans H: a basis.

□

A basis is as small and as large as possible.

- No vector can be removed from a basis $\mathcal{B}$ of H with the rest still spanning H: the removed vector would then be a combination of the others, and $\mathcal{B}$ would be dependent by (W1).
- No vector $\mathbf{u}$ of H can be added to $\mathcal{B}$ with the result still independent: $\mathbf{u}$ is a combination of the basis, so the enlarged set is dependent by (W1).

A basis for $\operatorname{Nul} A$: the vectors from the parametric vector form of $A\mathbf{x} = \mathbf{0}$, which span $\operatorname{Nul} A$ and are independent (W4).

Theorem (A basis for $\operatorname{Col} A$ (Lay 4.3, Thm. 6))

The pivot columns of a matrix A form a basis for $\operatorname{Col} A$.

The idea

Row reduction changes the columns, so $\operatorname{Col} A$ and the column space of an echelon form are usually different. But it does not change which combinations of the columns are zero, since those combinations are the solutions of $A\mathbf{x} = \mathbf{0}$. In the reduced echelon form the dependence relations among the columns can be read off, and the same relations hold among the columns of A.

Proof. 1. Let B be the reduced echelon form of A. A list of weights $\mathbf{x}$ satisfies $x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n = \mathbf{0}$ exactly when $A\mathbf{x} = \mathbf{0}$, and by (W3) exactly when $B\mathbf{x} = \mathbf{0}$, that is, $x_1\mathbf{b}_1 + \cdots + x_n\mathbf{b}_n = \mathbf{0}$. So the columns of A and of B satisfy the same linear relations.

2. The pivot columns of B are distinct standard vectors $\mathbf{e}_i$, so they are independent.

3. Each non-pivot column $\mathbf{b}_j$ of B is a combination of the pivot columns to its left. Reason: an entry $b_{ij} \neq 0$ is not to the left of the leading 1 of row i (the leading entry is the first nonzero one), and it is not in that pivot's column, since $\mathbf{b}_j$ is not a pivot column; so $b_{ij} \neq 0$ only for rows i whose pivot lies to the left of column j. The pivot column of row i is $\mathbf{e}_i$, so $\mathbf{b}_j = \sum_i b_{ij}\mathbf{e}_i$ is a combination of those pivot columns.

4. By step 1, the pivot columns of A are independent and each non-pivot column of A is the same combination of pivot columns of A. By the spanning set theorem (a), removing the non-pivot columns one at a time does not change $\operatorname{Col} A$. So the pivot columns of A span $\operatorname{Col} A$ and are independent: they are a basis of $\operatorname{Col} A$.

□

Theorem (Row spaces (Lay 4.3))

If A and B are row equivalent, then $\operatorname{Row} A = \operatorname{Row} B$. If B is in echelon form, the nonzero rows of B form a basis for $\operatorname{Row} A$.

Proof. 1. Each row operation replaces rows by combinations of rows, so the rows after the operation lie in the row space before it. Each operation can be undone by a row operation (W3), so the reverse inclusion holds too. So the row space never changes.

2. The nonzero rows of an echelon form B span $\operatorname{Row} B = \operatorname{Row} A$ (zero rows add nothing).

3. They are independent. In a relation $c_1\mathbf{r}_1 + \cdots + c_k\mathbf{r}_k = \mathbf{0}$ among the nonzero rows, look at the column of the leading entry of $\mathbf{r}_1$: every later row has 0 there, so that entry of the sum is c_1 times the leading entry, and $c_1 = 0$. Then look at the leading column of $\mathbf{r}_2$ in the same way to get $c_2 = 0$, and so on down the rows.

□

The question answered. Row reduce A:

$$A \xrightarrow[R_4 - 5R_1]{R_2 - 3R_1,\ R_3 - 2R_1} \begin{bmatrix} 1 & 4 & 0 & 2 & -1 \\ 0 & 0 & 1 & -1 & 8 \\ 0 & 0 & 1 & -1 & 4 \\ 0 & 0 & 2 & -2 & 13 \end{bmatrix} \xrightarrow[R_4 - 2R_2]{R_3 - R_2} \begin{bmatrix} 1 & 4 & 0 & 2 & -1 \\ 0 & 0 & 1 & -1 & 8 \\ 0 & 0 & 0 & 0 & -4 \\ 0 & 0 & 0 & 0 & -3 \end{bmatrix} \xrightarrow{R_4 - \frac{3}{4}R_3} \begin{bmatrix} 1 & 4 & 0 & 2 & -1 \\ 0 & 0 & 1 & -1 & 8 \\ 0 & 0 & 0 & 0 & -4 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

and then $-\frac{1}{4}R_3$, $R_2 - 8R_3$, $R_1 + R_3$ give the reduced echelon form

$$B = \begin{bmatrix} 1 & 4 & 0 & 2 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

1. The columns of B show $\mathbf{b}_2 = 4\mathbf{b}_1$ and $\mathbf{b}_4 = 2\mathbf{b}_1 - \mathbf{b}_3$, so columns 2 and 4 of A can be dropped: $\mathbf{a}_2 = 4\mathbf{a}_1$ and $\mathbf{a}_4 = 2\mathbf{a}_1 - \mathbf{a}_3$. The smallest spanning sets are bases.

2. $\operatorname{Col} A$: the pivot columns of A (not of B), $\mathbf{a}_1 = (1, 3, 2, 5)$, $\mathbf{a}_3 = (0, 1, 1, 2)$, $\mathbf{a}_5 = (-1, 5, 2, 8)$.

3. Nul A: x_2, x_4 are free, $x_1 = -4x_2 - 2x_4$, $x_3 = x_4$, $x_5 = 0$. Basis $(-4,1,0,0,0)$, $(-2,0,1,1,0)$.
4. Row A: the nonzero rows of B, $(1,4,0,2,0)$, $(0,0,1,-1,0)$, $(0,0,0,0,1)$.

Shortcuts

- **Bases for the three spaces from one row reduction**: Col A: pivot columns of A; Row A: nonzero rows of an echelon form; Nul A: the vectors from the parametric form. Reason: the three theorems above. Instance: the question.
- **Basis of a subspace given as a span**: put the vectors in as columns and keep the pivot columns. Reason: Theorem 6. Instance: for $\{(1,2),(2,4),(0,1)\}$ the pivot columns of the matrix with these columns are 1 and 3, so $(1,2)$, $(0,1)$ is a basis of the span ($\mathbb{R}^2$).
- **A basis of** Row A **made of rows of** A: the pivot columns of A^T tell which rows of A to keep. Reason: Row A = Col A^T and Theorem 6. Instance: for the matrix A below, $R_2 - R_1$ reduces A^T as shown, so the pivot columns of A^T are 1 and 3, and rows 1 and 3 of A, $(1,1)$ and $(0,1)$, form a basis of Row A.

$$A = \begin{bmatrix} 1 & 1 \\ 2 & 2 \\ 0 & 1 \end{bmatrix}, \qquad A^T = \begin{bmatrix} 1 & 2 & 0 \\ 1 & 2 & 1 \end{bmatrix} \xrightarrow{R_2 - R_1} \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

- **Polynomial subspaces**: write the conditions as equations on the coefficients, solve, and split by free coefficient. Reason: the coordinates of $a_0 + a_1 t + a_2 t^2$ are (a_0, a_1, a_2). Instance: $\{p \in \mathbb{P}_2 : p(0) = 0\}$ means $a_0 = 0$, so $p = a_1 t + a_2 t^2$, and $\{t, t^2\}$ is a basis (different degrees, so independent).

Traps

- The basis for Col A is the pivot columns *of* A. For the matrix below, the reduced echelon form has the column $(1,0)$, which is not even in Col $A = \text{Span}\{(1,2)\}$.

$$\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \xrightarrow{R_2 - 2R_1} \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}$$

- The basis for Row A is the nonzero rows *of the echelon form*, not the corresponding rows of A. For the matrix A of the shortcut "A basis of Row A made of rows of A", an echelon form has nonzero rows $(1,1)$, $(0,1)$ in rows 1 and 2, but rows 1 and 2 of A, $(1,1)$ and $(2,2)$, are dependent.

$$\begin{bmatrix} 1 & 1 \\ 2 & 2 \\ 0 & 1 \end{bmatrix} \xrightarrow[R_2 \leftrightarrow R_3]{R_2 - 2R_1} \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$$

- Row operations do not preserve dependence relations among the rows. The rows $\mathbf{r}_1 = (1,1)$ and $\mathbf{r}_2 = (2,2)$ satisfy $2\mathbf{r}_1 - \mathbf{r}_2 = \mathbf{0}$; after $R_2 - 2R_1$ the rows are $(1,1)$ and $(0,0)$, and $2(1,1) - (0,0) \neq \mathbf{0}$. (Relations among the *columns* are preserved, by step 1 of Theorem 6.)
- A basis is not unique: $\{(1,0),(0,1)\}$ and $\{(1,1),(1,-1)\}$ are both bases of $\mathbb{R}^2$.

§2.2 The dimension of a vector space (Lay 4.5)

The question. $\{(1,0),(0,1)\}$ and $\{(1,1),(1,-1)\}$ are both bases of $\mathbb{R}^2$. Must every basis of $\mathbb{R}^2$ have two vectors? Can a plane through $\mathbf{0}$ in $\mathbb{R}^3$ have a basis of three vectors? If all bases of a subspace have the same size, that number measures the subspace.

Theorem (More vectors than a basis are dependent (Lay 4.5, Thm. 9))
If a vector space V has a basis $\{\mathbf{b}_1, \dots, \mathbf{b}_n\}$, then any set in V containing more than n vectors is linearly dependent.

The idea

Write each of the vectors $\mathbf{u}_1, \dots, \mathbf{u}_p$ $(p > n)$ in terms of the basis. A dependence relation among the $\mathbf{u}$'s then comes from a homogeneous linear system with n equations (one per basis vector) and p unknowns (one weight per $\mathbf{u}_j$), and more unknowns than equations always give a nontrivial solution.

Proof. 1. Let $\mathbf{u}_1, \dots, \mathbf{u}_p$ be in V with $p > n$. Write $\mathbf{u}_j = a_{1j}\mathbf{b}_1 + \cdots + a_{nj}\mathbf{b}_n$, and let M be the $n \times p$ matrix with entries a_{ij}.

2. Since $p > n$, $M\mathbf{c} = \mathbf{0}$ has a nontrivial solution $\mathbf{c} = (c_1, \dots, c_p)$ (W2).

3. Then, regrouping by basis vector,

$$c_1\mathbf{u}_1 + \cdots + c_p\mathbf{u}_p = \sum_{j=1}^{p} c_j \sum_{i=1}^{n} a_{ij}\mathbf{b}_i = \sum_{i=1}^{n} \Big(\sum_{j=1}^{p} a_{ij}c_j\Big)\mathbf{b}_i = \sum_{i=1}^{n} (M\mathbf{c})_i\, \mathbf{b}_i = \mathbf{0},$$

with the c_j not all 0.

□

Theorem (All bases have the same size (Lay 4.5, Thm. 10))
If a vector space V has a basis of n vectors, then every basis of V has exactly n vectors.

Proof. 1. Let $\mathcal{B}$ be a basis of n vectors and $\mathcal{C}$ another basis, with m vectors.

2. $\mathcal{C}$ is independent, so by Theorem 9 applied to the basis $\mathcal{B}$, $m \le n$.

3. $\mathcal{B}$ is independent, so by Theorem 9 applied to the basis $\mathcal{C}$, $n \le m$. So $m = n$.

□

Definition. If V is spanned by a finite set, V is **finite-dimensional**, and the **dimension** $\dim V$ is the number of vectors in a basis of V; $\dim\{\mathbf{0}\} = 0$. Otherwise V is **infinite-dimensional**.

A finite-dimensional $V \neq \{\mathbf{0}\}$ has a basis by the spanning set theorem (b), and all its bases have the same size by Theorem 10, so $\dim V$ is well defined.

- $\dim \mathbb{R}^n = n$ (the standard basis) and $\dim \mathbb{P}_n = n + 1$ (the basis $1, t, \dots, t^n$).
- The space $\mathbb{P}$ of all polynomials is infinite-dimensional: a finite set of polynomials has a largest degree d, its span contains only polynomials of degree at most d, and t^{d+1} is not among them.
- **Bounds.** If p vectors span V, then $\dim V \le p$, because they contain a basis (spanning set theorem). If V contains p independent vectors, then $\dim V \ge p$, because otherwise Theorem 9 would make them dependent.

Worked example: dimension of a subspace given by parameters. Let $H = \{(a+2b,\ b-c,\ a+2c,\ 3a+6b)\}$ in $\mathbb{R}^4$, with a, b, c real.

1. Split by parameter: the vector is $a\mathbf{v}_1 + b\mathbf{v}_2 + c\mathbf{v}_3$ with $\mathbf{v}_1 = (1, 0, 1, 3)$, $\mathbf{v}_2 = (2, 1, 0, 6)$, $\mathbf{v}_3 = (0, -1, 2, 0)$, so $H = \mathrm{Span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$.
2. $\mathbf{v}_3 = 2\mathbf{v}_1 - \mathbf{v}_2$, so by the spanning set theorem $H = \mathrm{Span}\{\mathbf{v}_1, \mathbf{v}_2\}$. Neither of these is a multiple of the other (look at entries 2 and 3), so they are independent.
3. So $\{\mathbf{v}_1, \mathbf{v}_2\}$ is a basis and $\dim H = 2$, even though H was written with three parameters.

Theorem (Subspaces of a finite-dimensional space (Lay 4.5, Thm. 11))

Let H be a subspace of a finite-dimensional vector space V. Any linearly independent set in H can be extended, if necessary, to a basis for H. Also, H is finite-dimensional and $\dim H \leq \dim V$.

Proof. 1. If $H = \{\mathbf{0}\}$, then $\dim H = 0 \leq \dim V$. Otherwise let S be an independent set in H (if none is given, start with one nonzero vector of H).

2. **Assume S does not span H.** Choose $\mathbf{u}$ in H outside $\operatorname{Span} S$. Then $S \cup \{\mathbf{u}\}$ is independent: in a relation, the weight on $\mathbf{u}$ must be 0 (otherwise $\mathbf{u}$ would be a combination of S), and then all the other weights are 0 because S is independent.

3. Repeat step 2. Every set produced is independent in V, so by Theorem 9 it has at most $\dim V$ vectors. So the process stops, and it stops only when the set spans H: a basis of H with at most $\dim V$ vectors.

□

Theorem (The basis theorem (Lay 4.5, Thm. 12))

Let V be a p-dimensional vector space, $p \geq 1$. Any linearly independent set of exactly p vectors in V is a basis for V. Any set of exactly p vectors that spans V is a basis for V.

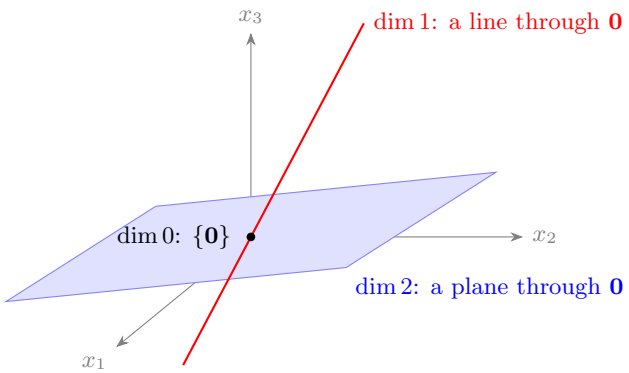
Proof. 1. **Assume S is independent with p vectors.** By Theorem 11 (with $H = V$), S extends to a basis of V, which has exactly p vectors (Theorem 10). So nothing was added, and S is a basis.

2. **Assume S spans V with p vectors.** By the spanning set theorem, S contains a basis of V, which has exactly p vectors. So nothing was removed, and S is a basis.

□

Subspaces of $\mathbb{R}^3$ by dimension. A subspace H of $\mathbb{R}^3$ has $\dim H \leq 3$ by Theorem 11.

- $\dim H = 0$: $H = \{\mathbf{0}\}$.
- $\dim H = 1$: $H = \operatorname{Span}\{\mathbf{v}\}$ with $\mathbf{v} \neq \mathbf{0}$, a line through $\mathbf{0}$.
- $\dim H = 2$: $H = \operatorname{Span}\{\mathbf{u}, \mathbf{v}\}$ with $\mathbf{u}, \mathbf{v}$ independent, a plane through $\mathbf{0}$.
- $\dim H = 3$: a basis of H is 3 independent vectors in $\mathbb{R}^3$, which by the basis theorem is a basis of $\mathbb{R}^3$; so $H = \mathbb{R}^3$.



The proper subspaces of $\mathbb{R}^3$: the origin, lines through the origin, and planes through the origin. The only 3-dimensional subspace is $\mathbb{R}^3$ itself.

The question answered. Every basis of $\mathbb{R}^2$ has 2 vectors (Theorem 10). A plane through $\mathbf{0}$ in $\mathbb{R}^3$ is $\operatorname{Span}\{\mathbf{u}, \mathbf{v}\}$ with $\mathbf{u}, \mathbf{v}$ independent, so it has dimension 2; by Theorem 9 any three vectors in it are dependent, so it has no basis of 3 vectors.

Shortcuts

- **Is this set a basis of $\mathbb{R}^n$?** With exactly n vectors, check independence only (or $\det \neq 0$). With fewer than n: no, since by the bounds above they cannot span. With more than n: no, since they are dependent

(Theorem 9). Reason: the basis theorem. Instance: $(1,2)$, $(3,4)$ have $\det = 4-6 \neq 0$, so they are a basis of $\mathbb{R}^2$.

- **Extending to a basis of $\mathbb{R}^n$**: row reduce $[\,\mathbf{v}_1 \cdots \mathbf{v}_k\ \mathbf{e}_1 \cdots \mathbf{e}_n\,]$ and keep the pivot columns. Reason: the columns span $\mathbb{R}^n$ (they include the $\mathbf{e}_i$), and by step 3 of Theorem 6 a column fails to be a pivot column only if it is a combination of earlier columns, which never happens for the independent $\mathbf{v}$'s. Instance: for $(1,1,0)$, row reduction gives pivot columns $1,2,4$, so the basis is $(1,1,0)$, $\mathbf{e}_1$, $\mathbf{e}_3$.

$$\begin{bmatrix}1&1&0&0\\1&0&1&0\\0&0&0&1\end{bmatrix} \xrightarrow{R_2-R_1} \begin{bmatrix}1&1&0&0\\0&-1&1&0\\0&0&0&1\end{bmatrix}$$

- **Dimension of a span**: the number of pivot columns of the matrix with those columns. Reason: Theorem 6. Instance: the worked example has $\dim H = 2$.

Traps

- The basis theorem needs the right count: two independent vectors in $\mathbb{R}^3$, such as $\mathbf{e}_1, \mathbf{e}_2$, are not a basis of $\mathbb{R}^3$.
- $\dim \mathbb{R}^2 = 2$, but $\mathbb{R}^2$ is not a subspace of $\mathbb{R}^3$; a plane through $\mathbf{0}$ in $\mathbb{R}^3$ is a 2-dimensional subspace of $\mathbb{R}^3$.
- A dependent set gives no bound on the dimension: $\{\mathbf{0}, \mathbf{e}_1\}$ is a dependent set of 2 vectors in $\mathbb{R}^3$, which has dimension 3.
- The number of parameters is not the dimension: the worked example uses three parameters and has dimension 2.

§2.3 Rank (Lay 4.5)

The question. A 7×9 matrix has a two-dimensional null space. Without seeing the matrix, what is the dimension of its column space, and is $\operatorname{Col} A$ all of $\mathbb{R}^7$?

- $\dim \operatorname{Nul} A$ is the number of free variables, because the parametric vector form gives a basis of $\operatorname{Nul} A$ with one vector per free variable (W4).
- $\dim \operatorname{Col} A$ is the number of pivot columns (Theorem 6).
- $\dim \operatorname{Row} A$ is the number of pivots, the number of nonzero rows of an echelon form (row space theorem).

Definition. The **rank** of A is $\operatorname{rank} A = \dim \operatorname{Col} A$. The **nullity** is $\dim \operatorname{Nul} A$.

By the three items above, $\operatorname{rank} A = \dim \operatorname{Row} A$, and since $\operatorname{Row} A = \operatorname{Col} A^T$, also $\operatorname{rank} A = \operatorname{rank} A^T$.

Theorem (The rank theorem (Lay 4.5))
For an $m \times n$ matrix A, $\operatorname{rank} A + \dim \operatorname{Nul} A = n$.

Proof. 1. Each of the n columns of A is either a pivot column or the column of a free variable, and not both.

2. The pivot columns number $\operatorname{rank} A$ (Theorem 6) and the free variables number $\dim \operatorname{Nul} A$ (the parametric vector form gives a basis of $\operatorname{Nul} A$ with one vector per free variable, W4). So the two numbers add up to n.

□

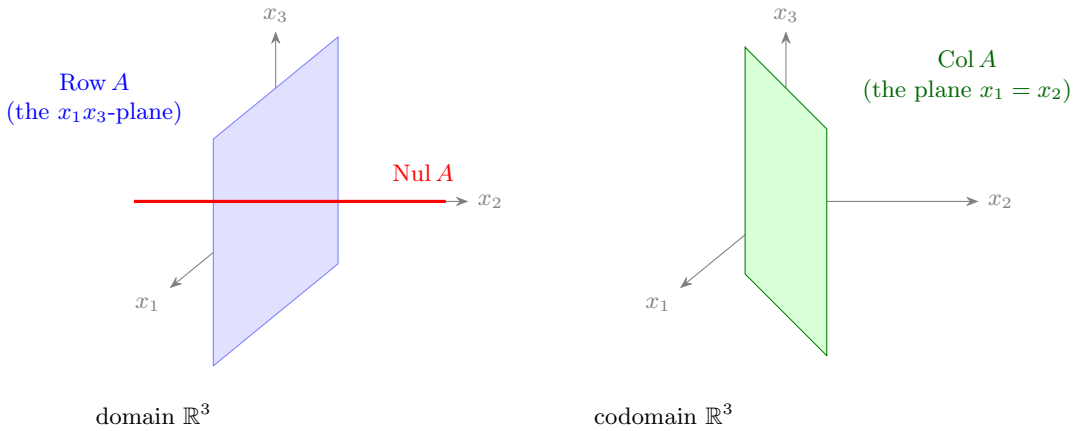
The question answered. By the rank theorem, $\operatorname{rank} A = 9 - 2 = 7$. So $\operatorname{Col} A$ is a 7-dimensional subspace of $\mathbb{R}^7$; a basis of it is 7 independent vectors in $\mathbb{R}^7$, hence a basis of $\mathbb{R}^7$ (basis theorem), and $\operatorname{Col} A = \mathbb{R}^7$.

Worked example: consistency from dimensions. A homogeneous system of 50 equations in 54 unknowns has a solution set spanned by 4 independent vectors. Is every system with the same coefficients and any right-hand side consistent? The coefficient matrix A is 50×54 with $\dim \operatorname{Nul} A = 4$, so $\operatorname{rank} A = 54 - 4 = 50$, and as above $\operatorname{Col} A = \mathbb{R}^{50}$. So $A\mathbf{x} = \mathbf{b}$ is consistent for every $\mathbf{b}$ (W2).

The invertible matrix theorem, continued (Lay 4.5). For an $n \times n$ matrix A, each of the following is equivalent to A being invertible: (m) the columns of A form a basis of $\mathbb{R}^n$; (n) $\operatorname{Col} A = \mathbb{R}^n$; (o) $\dim \operatorname{Col} A = n$; (p) $\operatorname{rank} A = n$; (q) $\operatorname{Nul} A = \{\mathbf{0}\}$; (r) $\dim \operatorname{Nul} A = 0$. Reason: (m) is "independent and spanning"; (n), (o), (p) each say there are n pivot columns, using the basis theorem for (o)⇒(n); (q) is "$A\mathbf{x} = \mathbf{0}$ only trivially"; (r) is (q) restated; (p) $\iff$ (r) is the rank theorem.

The three spaces in a picture. The figure shows $\operatorname{Row} A$, $\operatorname{Nul} A$ and $\operatorname{Col} A$ for

$$A = \begin{bmatrix} 3 & 0 & -1 \\ 3 & 0 & -1 \\ 4 & 0 & 5 \end{bmatrix}.$$



The three spaces of the matrix A above. Rank 2: $\operatorname{Row} A$ and $\operatorname{Col} A$ are planes, and $\operatorname{Nul} A$ is a line ($2 + 1 = 3$ columns). $\operatorname{Nul} A$ is the x_2-axis, because the equations $3x_1 - x_3 = 0$ and $4x_1 + 5x_3 = 0$ force $x_1 = x_3 = 0$.

Shortcuts

- **Counting**: $\operatorname{rank} A$ is the number of pivots, which equals $\dim \operatorname{Col} A$ and $\dim \operatorname{Row} A$; and $\dim \operatorname{Nul} A = n - \operatorname{rank} A$. Reason: the rank theorem. Instance: the question.
- **Rank bounds**: $\operatorname{rank} A \leq \min(m, n)$, since there is at most one pivot per row and per column (W5). Instance: a 6×9 matrix has rank at most 6, so its null space has dimension at least $9 - 6 = 3$.
- **Full rank m means onto**: if $\operatorname{rank} A = m$ (the number of rows), then $\operatorname{Col} A = \mathbb{R}^m$ and $A\mathbf{x} = \mathbf{b}$ is always consistent. Reason: basis theorem. Instance: the 50×54 example.

Traps

- The rank theorem has n (the number of columns), not m: for the question's 7×9 matrix, $7 + 2 = 9$.
- $\dim \operatorname{Row} A = \dim \operatorname{Col} A$, but the spaces themselves differ in general: for the matrix below, $\operatorname{Row} A = \operatorname{Span}\{(0, 1)\}$ and $\operatorname{Col} A = \operatorname{Span}\{(1, 0)\}$.

$$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

- $\operatorname{Nul} A$ and $\operatorname{Col} A$ live in different spaces when $m \neq n$: for a 3×7 matrix of rank 3, $\operatorname{Nul} A$ is a 4-dimensional subspace of $\mathbb{R}^7$, not $\mathbb{R}^4$.

§3 Homework this week

HW5, Section 4.3, and HW6, Section 4.5.

Section 4.3.

- **#3, #5** (bases of $\mathbb{R}^3$?). Row reduce the matrix with these columns: a pivot in every column gives independence, a pivot in every row gives spanning; a set containing $\mathbf{0}$ is dependent.
- **#13** (bases for $\operatorname{Col} A$, $\operatorname{Row} A$, $\operatorname{Nul} A$ from $A \sim B$). Pivot columns of A; nonzero rows of B; the parametric form (as in the question).
- **#15** (basis of a span). Vectors as columns; keep the pivot columns.
- **#20** (a given relation). Drop a vector whose weight in the relation is nonzero (spanning set theorem); the remaining two are not multiples of each other.
- **#23, #24, #26, #29, #31** (true/false). A basis needs independence and spanning; the spanning set theorem (b); a basis is a largest independent set; $A\mathbf{x} = \mathbf{0}$ and $B\mathbf{x} = \mathbf{0}$ have the same solutions; the trap on row relations.
- **#41** (dependent images). Apply T to a nontrivial relation and use linearity.
- **#44** (polynomials p_1, p_2, p_3). Find the relation among them, then the spanning set theorem.

Section 4.5.

- **#1, #3** (dimension of a subspace given by parameters). Split by parameter, remove redundant vectors (worked example).
- **#9, #11** (dimension of a span; of $\operatorname{Nul} A$, $\operatorname{Col} A$, $\operatorname{Row} A$). Count pivot columns and free variables.
- **#17, #18, #22, #24, #25** (true/false). Theorem 6; nullity is the number of free variables; both ranks are the number of pivots; the rank theorem; the bound from a spanning set.
- **#27** (Hermite polynomials). Different degrees, so independent; then the basis theorem in $\mathbb{P}_3$.
- **#33, #35** (counting). The rank theorem and $\operatorname{rank} A = \operatorname{rank} A^T$; a 4-dimensional subspace of $\mathbb{R}^4$ is $\mathbb{R}^4$, and $\operatorname{Nul} A$ lives in $\mathbb{R}^n$.
- **#44, #45** (true/false on bounds). A dependent set gives no bound (third trap of 4.5); extend an independent set (Theorem 11).

§4 Problem set

P1. For the matrix below, find bases for $\operatorname{Col} A$, $\operatorname{Row} A$ and $\operatorname{Nul} A$, and check the rank theorem.

$$A = \begin{bmatrix} 1 & 2 & -1 & 3 \\ 2 & 4 & 1 & 0 \\ 3 & 6 & 0 & 3 \end{bmatrix}$$

P2. Is $\{(1, 0, -1), (2, 1, 0), (1, 3, 4)\}$ a basis of $\mathbb{R}^3$?

P3. Find a basis for, and the dimension of, $H = \operatorname{Span}\{(1, 0, 2), (2, 1, 5), (0, -1, -1)\}$.

P4. Find a basis for $\{p \in \mathbb{P}_3 : p(1) = 0\}$ and its dimension.

P5. (a) A is 5×8 with rank 5. What is $\dim \operatorname{Nul} A$? Is $\operatorname{Col} A = \mathbb{R}^5$? (b) Can a 4×6 matrix have a one-dimensional null space?

P6. Extend $\{(1, 2, 0), (0, 1, 3)\}$ to a basis of $\mathbb{R}^3$.

P7. True or false, with a reason. (a) If $\dim V = 3$, any 4 vectors in V are dependent. (b) If $\dim V = 3$, any 2 vectors in V span a 2-dimensional subspace. (c) The number of pivot columns of A equals $\dim \operatorname{Row} A$. (d) If A is 3×3 and $\dim \operatorname{Nul} A = 1$, then $\det A = 0$.